

A Proposal for Multi-modal Emotionally Intelligent System

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Introduction

Artificial Intelligence (AI) has progressed remarkably in areas such as logical reasoning, pattern recognition, and automation. As AI becomes increasingly integrated into our daily lives, its ability to recognize and respond to human emotions will be essential for meaningful interactions. Emotional Intelligence (EI) in AI is therefore not just a desirable feature, but a necessary next step.

Problem Statement

Present-day AI systems are capable of exchanging information and simulating rational dialogue, but frequently lack the capacity to recognize or react sympathetically to human emotions. For example, current mental health chatbots can provide scripted responses but fail to detect when a user is genuinely distressed versus merely reporting a bad day. This lessens the depth and efficacy of communication, particularly in delicate situations like stress management, e-learning, telemedicine, or mental health assistance. Without emotional intelligence, AI systems may struggle to maintain long-term, trust-based interactions with users.. This system could support applications such as mental health monitoring, conversational agents, and digital well-being systems.

Deep Learning Task/ Motivation & Objectives

In this project, a machine learning model will be trained to classify the emotions expressed in facial expressions and text data. This project aims to embed emotional intelligence into a deep learning system by training the model with available datasets. In addition to using the datasets, by reviewing existing literature, this work will seek to develop a data mining project that can train a model to perform prediction of emotions classification on unseen data.

Deep learning is uniquely suited for this task because:

- CNNs excel at extracting emotional features from facial expressions
- Transformers/RNNs capture contextual and sentimental nuances in text
- Multi-modal deep learning architectures can fuse these complementary signals for more robust emotion recognition than any single modality.

The objectives include, but are not limited to, the following:

- Create a mapping of the emotional states for limited classification labels.
- Determine the most important techniques in the use case, such as multimodal emotion recognition, sentiment analysis, affective computing, and natural language processing.
- Develop models capable of classifying human emotions from facial expressions and text.

Data

The datasets used for training and evaluating the multi-modal emotionally intelligent system for this project will come from a Kaggle dataset called “Emotions Dataset” for text. And for facial images, the dataset is “Facial Emotional Recognition Image Dataset”. We aim to integrate other datasets to make the model more robust and useful in various scenarios as the project progresses.

DATASET 1: Text-Based Emotions Dataset

Source: Kaggle “Emotions Dataset” by Bhavik Jikadara

<https://www.kaggle.com/datasets/bhavikjikadara/emotions-dataset/data>

This dataset is designed to help research in natural language preprocessing and emotion analysis. It contains a diverse collection of text samples drawn from Twitter messages, each labelled with the predominant emotion it conveys. Each entry consists of a text segment, a tweet, and a numeric label for the corresponding emotion. This dataset is well-suited for training transformer models.

Key characteristics:

Attribute	Details
Size	About 400k+ rows
Format	CSV file (emotions.csv) with ‘text’ and ‘label’ columns
Source Domain	English Twitter messages
Emotion Classes	6 categories: sadness (0), joy (1), love (2), anger (3), fear (4), surprise (5)
Task Suitability	Sentiment analysis, emotion classification, textual/linguistic analysis

DATASET 2: Facial Emotion Recognition Image Dataset

Source: Kaggle “Facial Emotion Recognition Image Dataset” by Kovenko and Shevchuk (2021)

<https://www.kaggle.com/datasets/sujaykapadnis/emotion-recognition-dataset>

This image dataset was constructed by scraping social networks, YouTube videos, and existing datasets. It contains RGB cropped face images organized by emotion class. It provides high visual diversity due to its multi-source collection strategy.

Key characteristics:

Attribute	Details
Size	About 15,500 unique values/images
Image Format	RGB, 48x48 pixels, cropped face images
File Organization	Dataset.zip: folders per emotion class; data.csv: image paths and labels
Emotion Classes	6 classes: happy, angry, sad, neutral, surprise, ahogado
Data Sources	Facebook, instagram, youtube, IMDB, affectnet
Task Suitability	Facial emotion recognition, CNN-based image classification

Because the datasets are not aligned across modalities, the system will initially train separate unimodal models for text and image emotion classification. Later, a fusion model will combine their prediction outputs to simulate multimodal inference. Synthetic pairing based on matching emotion labels may be explored to evaluate early fusion strategies.

Proposed Methodology

As a baseline, we will implement a simple, fully connected shallow network which will provide a reference for evaluating the performance of our deep learning and multi-modal models. The image emotion classifier will be implemented using a convolutional neural network (CNN) architecture in PyTorch, while text emotion classification will be implemented using a transformer-based model. Multi-modal fusion techniques will then be used to integrate the image and text representations for more comprehensive emotion classification.

Evaluation Plan

The datasets will be divided into training, validation, and testing sets (e.g., 70/15/15 split) to ensure unbiased evaluation. We will evaluate our models using metrics such as accuracy, a weighted F1-Score (class imbalance), precision/recall per class (which emotions are confused), and a confusion matrix (visualize misclassification patterns). We will also compare our models. We will have model types like baselines (SVM, random forest, shallow CNN), unimodal (image), unimodal (text), and multimodal where we compare within modality performance, across modality performance, fusion gains, and computational efficiency.

Preliminary Results and Baseline Solutions

Here, we will describe the preliminary data preprocessing and modeling work done for the text modality. Two approaches were developed and evaluated: (1) a classical machine learning baseline using TF-IDF features with ensemble and linear models, and (2) a deep learning approach using a pre-trained DistilBEERT transformer. Both approaches operate on the text emotions dataset and provide groundwork for the full multi-modal pipeline.

APPROACH 1: DL_text_classification.ipynb

This approach implements a complete data pre-processing and feature extraction pipeline with some classical machine learning models to serve as the required baseline for comparison against deep learning models.

Data preprocessing steps:

- Unicode normalization: text converted to ASCII using the unidecode library to remove special characters.
- All text converted to lowercase for consistency.
- Curly quote normalization: typographic quotes replaced with standard ASCII equivalents.
- Digit removal: Numerical characters removed using regex substitution.
- Whitespace normalization: multiple consecutive spaces -> a single space and leading/trailing whitespace trimmed.

Feature extraction:

Cleaned text was vectorized using Term Frequency-Inverse Document Frequency with English stop-word removal and vocabulary cap of 5000 features. A standard 80/20 train-text split was applied with stratification to preserve class balance.

Models trained and results:

Model	Train Accuracy	Test Accuracy	Training Time
Random Forest	0.9563	0.8561	About 1099 sec
Logistic Regression	0.9113	0.8969	About 10 sec

From these results, logistic regression achieved the best text accuracy (89.69%) while being faster to train than ensemble methods. These results establish the performance floor that the deep learning model must exceed.

APPROACH 2: text_emotion.py

This approach fine-tunes a pre-trained DistilBERT model for the 6-class emotion classification. DistilBERT was chosen for this task because it's a lightweight distilled version of BERT that retains strong language understanding while being significantly faster and smaller.

Implementation details:

- Model: DistilBertForSequenceClassification (6 output labels)
- Tokenizer: DistilBertTokenizer, max length 128 tokens, padding & truncation
- Optimizer: AdamW with learning rate 2e-5
- Loss function: CrossEntropyLoss
- Batch size: 64
- Epochs: 10
- Training subset: 10,000 samples (full dataset run planned)
- Hardware: GPU

Training progress (10,000 sample subset):

EPOCH	AVG. LOSS
1	1.0536
2	0.3190
3	0.1645
4	0.1162
5	0.0849
6	0.0718
7	0.0557
8	0.0418
9	0.0303
10	0.0307

This model converged steadily, with loss dropping from 1.05 in epoch 1 to 0.031 by epoch 10. Test accuracy on the 2000 sample held out set reached 91.8%, outperforming the classical baselines. Training on the full dataset (about 393,000 samples) is planned, which is expected to further improve accuracy and generalization when done.

APPROACH COMPARISON

Criterion	Classical ML (.ipynb)	DistilBERT Transformer (.py)
Best Test Accuracy	89.69% (logistic regr.)	91.8% (subset)
Preprocessing	Explicit cleaning pipeline	Tokenizer handles normalization
Feature Extraction	TF-IDF (sparse, no context)	Contextual embeddings (dense)
Scalability	Full dataset used	Subset only; full run planned
Role in Project	Baseline models	Primary text classifier
Visualization	EDA plots	Training loss per epoch

The DistilBERT script is the primary deep learning solution for the text modality and will be further developed with full-dataset training, additional evaluation metrics like F1-score, and model saving. The next stage will involve the image modality CNN implementation.

Model Training and Testing

For facial recognition, a Deep Learning model with CNN, PyTorch was prepared and trained with split as 70:15:15 for training, validating and testing datasets. We tried different approaches and in different servers.

Loss function:



The plot below shows the training loss decreasing from 1.6333 to 1.1815 over 25 epochs, indicating the model was successfully learning. However, the accuracy was not in par with our expectations. Even class imbalance handling dropped the accuracy rather than increasing. Thus, we'll be working on improving performance.

Class	Precision	Recall	F1-Score	Support
Ahegao	0.58	0.90	0.71	181
Angry	0.00	0.00	0.00	197

Happy	0.84	0.87	0.85	561
Neutral	0.49	0.81	0.61	604
Sad	0.40	0.28	0.33	590
Surprise	0.32	0.08	0.13	185
Accuracy	-	-	0.57	2318
Macro Avg	0.44	0.49	0.44	2318
Weighted Avg	0.50	0.57	0.52	2318

Expected Outcomes

The CNN and transformer models are expected to achieve reliable emotion classification performance within their respective modalities. The multimodal model is expected to outperform unimodal models by leveraging complementary information from text and facial expressions. Potential challenges include limited data for certain emotions, risk of overfitting from limited data, and the complexity of implementing multi-modal fusion techniques to combine text and image recognition.

Schedule

The following table includes a tentative schedule for this project:

<u>Task Name</u>	<u>Start Date</u>	<u>Due Date</u>
1. Project proposal	2/3/2026	2/20/26
2. Data collection	2/21/26	3/6/2026
3. Data preprocessing & exploration	3/7/2026	3/20/26
4. Application of data mining algorithms	3/21/2026	4/3/2026
5. Comparison of algorithms through experiments	4/4/2026	4/17/2026
6. Preparation of report and presentation	4/18/26	4/30/2026
7. Project presentation	5/1/2026	5/1/2026

References

1. Emotions dataset
<https://www.kaggle.com/datasets/bhavikjikadara/emotions-dataset/data>
2. Facial Emotions recognition Image dataset
<https://www.kaggle.com/datasets/sujaykapadnis/emotion-recognition-dataset>
3. PyTorch
<https://pytorch.org/>